MOHAMED HAZEM ELALFY

AI Engineer
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SUMMARY

AI Engineering graduate from Mansoura University. Built machine learning and computer vision projects, a RAG Telegram chatbot, and an AI assistant for the IT support team during an internship at Elsewedy Electric. Completed the five-year engineering program in four years.

TECHNICAL SKILLS

Programming & data: Python, SQL, Git/GitHub, Pandas, NumPy, Jupyter
Machine learning: Scikit-learn, PyTorch, TensorFlow/Keras, classification, CNNs, model evaluation
LLM applications: Gemini and Groq APIs, RAG, text embeddings, cosine-similarity retrieval, prompt engineering
Applications & vision: Flask, REST APIs, webhooks, OpenCV, YOLOv8, MediaPipe; currently learning automation and AI agents

EXPERIENCE

Elsewedy Electric | AI Engineering Intern - IT Department | Aug-Sep 2026

- Developed AI tools for the IT support team within the IT Infrastructure & Operations department. Worked with two teammates to build SWE IT Assist, combining guided troubleshooting with a multilingual Gemini assistant.
- Implemented automatic ticket escalation when guided fixes and AI assistance could not resolve an issue; contributed to meeting-room booking using Python, PHP, JavaScript, and the Gemini API.

CodeAlpha | Machine Learning Intern (Remote) | Jul 2026

- Completed three classification projects: credit scoring, heart disease prediction, and handwritten digit recognition.
- Compared Logistic Regression, Decision Tree, and Random Forest on 1,000 simulated credit applications; Logistic Regression achieved 0.846 ROC-AUC on a 200-sample holdout set.
- Compared Logistic Regression, Random Forest, and SVM for heart disease prediction; trained a TensorFlow/Keras CNN for MNIST digit recognition with augmentation and training callbacks.

SELECTED PROJECTS

LLM Assistant with RAG | Python, Gemini, Groq, Render | 2026

- Built a Telegram chatbot that retrieves information from a company knowledge file using Gemini embeddings and cosine similarity; wrote the retrieval pipeline from scratch.
- Integrated the Groq API for response generation and deployed the application in webhook mode on Render.

Signa - AI Sign Language Creator Platform | MediaPipe, TensorFlow, Flask | 2026

- Developed the AI component in a seven-person team. Trained two gesture classifiers and connected gesture recognition, subtitles, text-to-speech, and video saving to a web interface using Flask.

Safe Eye - Real-Time Surveillance | PyTorch, YOLOv8, MViT, BiLSTM | 2025

- Led model selection and integration in a seven-person team. Combined fire, violence, and suspicious-behavior detection in one multi-threaded live-video pipeline.

Smart Home with Face Recognition | Python, OpenCV, Arduino | 2024

- Trained an LBPH face recognizer on locally collected data and connected recognition decisions to Arduino gate control.

EDUCATION

B.Eng. in Artificial Intelligence Engineering, Mansoura University | Sep 2022-Jun 2026
GPA: 3.15/4.0 | Completed in four years (standard program: five years) | Graduation project: Signa, A+

TRAINING & LANGUAGES

AMIT Learning - ML & AI Training (265+ hours) | Mansoura University - AI & Computer Vision Program
Arabic: Native | English: Upper-intermediate